

Chapter One

Network Troubleshooting Fundamentals

Troubleshooting is a systematic approach to solving a problem. The goal of troubleshooting is to determine why something does not work as expected and explain how to resolve the problem.

Troubleshooting network problems can be a frustrating experience, whether you're dealing with one PC or the entire ball of wax. This is because today's networks are so complicated that the point of failure could be virtually anywhere. Worse yet, your network could contain multiple points of failure, resulting in confusing symptoms that are hard to diagnose.

Troubleshooting Steps

In the Network+ troubleshooting model, there are eight steps:

1. Establish symptoms.
2. Identify the affected area.
3. Establish what has changed.
4. Select the most probable cause.
5. Implement a solution.
6. Test the result.
7. Recognize the potential effects of the solution.
8. Document the solution.

To facilitate our discussion of the troubleshooting steps, let's assume that a user has called you, the network administrator, to complain about not being able to connect to the Internet.

Step 1: Establish Symptoms

Obviously, if you can't identify a problem, you can't begin to solve it. Typically, you need to ask some questions to begin to clarify exactly what is happening. In our example, we should ask the user the following:

- Which part of the Internet can't you access?
- A particular website? A particular address? Any website?
- Can you use your web browser?

We find out that the user cannot access the corporate intranet or get to any sites on the Internet. He can, however, use his web browser to access the corporate FTP site, which he has bookmarked (by IP address 10.0.0.2). We can, therefore, rule out the web browser as the source of the problem.

Step 2: Identify the Affected Area

Computers and networks are fickle; they can work fine for months, suddenly malfunction horribly, and then continue to work fine for several more months, never again exhibiting that particular problem. And that's why it's important to be able to reproduce the problem and identify the affected area. Identifying the affected area narrows down what you have to troubleshoot.

One of your goals is to make problems easier to troubleshoot and, thus, get users working again as soon as possible. Therefore, the best advice you can give when training users is that when something isn't working, try it again and then write down exactly what is and is not happening. Most users' knee-jerk reaction is to call you immediately when they experience a problem. This isn't necessarily the best thing to do, because your response is most likely, "What were you doing when the problem occurred?" And most users don't know precisely what they were doing at the computer because they were primarily trying to get their job done. Therefore, if you train users to reproduce the problem first, they'll be able to give you the information you need to start troubleshooting it.

In our example, we find out that when the user tries to access the corporate intranet, he gets the following error message:

Step 3: Establish What Has Changed

If you can reproduce the problem, your next step is to attempt to determine the cause by determining what has changed. Drawing on your knowledge of networking, you might ask yourself and your user questions such as the following:

Were you ever able to do this? If not, then maybe this is not an operation the hardware or software is designed to do. You can inform the user that the system won't do the operation (or that she may need additional hardware or software to do it).

If so, when did you become unable to do it? If the computer was able to do the operation and then suddenly could not, the conditions that surround this change become extremely important. You may be able to discover the cause of the problem if you know what happened immediately before the change. It is likely that the cause of the problem is related to the conditions surrounding the change.

Has anything changed since you were last able to do this? This question can give you insight into a possible source for the problem. Most often, the thing that changed before the problem started is the source of the problem. When you ask this question of a user, the answer is typically that nothing has changed, so you might need to rephrase it. For example, you can try asking, "Did anyone add anything to your computer?" or "Are you doing anything differently from the way you normally proceed?"

Were any error messages displayed? This is one of the best indicators of the cause of a problem. Error messages are designed by programmers to help them determine what aspect of a computer system is not functioning correctly. These error messages are sometimes clear, such as "Disk Full" (indicating that the disk cannot store any more files on it because it is full). Or they can be cryptic, such as "A random bit has been flipped in the I/O subsystem of memory junction 44FA380h" (this

is a fictitious error, but you may encounter those just as complex). If you get a cryptic error message, you can go to the software or hardware vendor's support website and usually get a translation of the "programmers" of the error message into English.

Are other people experiencing this problem? This is one question you must ask yourself. That way you might be able to narrow down the problem to a specific item that may be causing the problem. Try to duplicate the problem yourself from your own workstation. If you can't duplicate the problem on another workstation, it may be related to only one user or group of users (or possibly their workstations). If more than one user is experiencing this problem, you may know this already because several people will be calling in with the same problem.

Is the problem always the same? Generally speaking, when problems crop up, they are almost always the same problem each time they occur. But their symptoms may change ever so slightly as conditions surrounding them change. A related question is, "If you do x, does the problem get better or worse?" For example, you might ask a user, "If you use a different file, does the problem get better or worse?" If the symptoms become less severe, it might indicate that the problem is related to the original file being used.

These are just a few of the questions you can use to isolate the cause of the problem.

In our example, we find out that the problem is unique to one user, indicating that the problem is specific to his workstation. When we watch him as he attempts to reproduce the problem, we notice that he is typing the address correctly. The error message leads us to believe that the problem has something to do with DNS (Domain Name Service) lookups on his workstation.

Step 4: Select the Most Probable Cause

After you observe the problem and isolate the cause, your next step is to select the most probable cause for the problem. Trust me, this gets easier with time and experience.

You must come up with at least one possible cause, even though it may not be correct. And you don't always have to come up with it yourself. Someone else in the group may have the answer. Also, don't forget to check online sources and vendor documentation.

In our example, we determined earlier that the cause was improperly configured DNS lookup on the workstation. The correction, then, is to reconfigure DNS on the workstation.

Step 5: Implement a Solution

In this step, you implement the solution. In our example, we need to reconfigure DNS on the workstation by following these steps:

1. Choose Start ➤ Settings ➤ Control Panel ➤ Network to open the Network dialog box.
2. Click the TCP/IP binding for your network card (indicated by TCP/IP ➤ *name of network card*).
3. Click Properties to open the TCP/IP Properties dialog box for that binding.

4. Click the DNS Configuration tab.

Step 6: Test the Result

Now that you have made the changes, you must test your solution to see if it solves the problem. In our example, we'd ask the user to try to access the intranet (since that was the problem reported). In general terms, ask the user to repeat the operation that previously did not work. If it works, great! The problem is solved. If it doesn't, try the operation yourself.

If the problem isn't solved, you may have to go back to step 4, select a new possible cause, and redo steps 5 and 6. But it is important to make note of what worked and what didn't so that you don't make the same mistakes twice.

Step 7: Recognize the Potential Effects of the Solution

The fundamental flaw of any network technician is the ability of the technician to solve only the one problem and not realize what other problems that solution may cause. It is possible that the solution may be worse than the problem. As the saying goes, "Sometimes the cure is worse than the disease."

Before fully implementing the solution to a problem, make sure you are completely aware of the potential effects of the solution and the other problems it may cause. If it causes more problems than it fixes, the solution isn't probably the best solution for the problem.

Step 8: Document the Solution

Network documentation is very important. You'll definitely want to document problems and solutions so that you have the information at hand when a similar problem arises in the future. With documented solutions to documented problems, you can assemble your own database of information that you can use to troubleshoot other problems. Be sure to include information such as the following:

- A description of the conditions surrounding the problem
- The NOS version, the software version, the type of computer, and the type of NIC
- Whether you were able to reproduce the problem
- The solutions you tried
- The ultimate solution

Hardware Troubleshooting Tools

In addition to manufacturer-provided troubleshooting tools, there are a few hardware devices we can use to troubleshoot the network. These are actual devices that you can use during the troubleshooting process. Some devices have easily recognizable functions; others are more obscure. Four of the most popular hardware tools are:

- A crossover cable

- A hardware loopback
- A tone generator
- A tone locator

Software Troubleshooting Tools

In addition to these hardware troubleshooting tools, you can use software programs to gain information about the current health and state of the network. These tools fall into two main categories:

- Protocol analyzers
- Performance-monitoring tools

We use the term *network software diagnostics* to refer to these tools.

Prioritize Your Problems

It is unlikely that as a network administrator or technician, you will receive problem calls one at a time. Typically, when you receive one call, you already have three people waiting for service. For this reason, you must learn to prioritize.

You start this process by asking some basic questions of the person reporting the problem so that you can determine its severity. If the current problem is minor and you have two more serious problems already facing you, your priorities are obvious.

You establish priorities to ensure that you spend your time wisely. The order in which you attempt to solve your networking problems, from highest priority to lowest, might look something like this:

- Total network failure (affects everyone)
- Partial network failure (affects small groups of users)
- Small network failure (affects a small, single group of users)
- Total workstation failure (single user can't work at all)
- Partial workstation failure (single user can't do most tasks)
- Minor issue (single user has problems that crop up now and again)

Mitigating circumstances can, of course, change the order of this list. For example, if the president of the company can't retrieve her e-mail, you'd take the express elevator to her office as soon as you hang up from the call. Also, a minor, persistent problem might move up the ladder.

Remember also that some simple problems may take more effort than larger problems. You may be able to bring up a crashed server in a matter of minutes, but a user who doesn't know how to make columns line up in Microsoft Word may take up to an hour or longer to train. The latter of these problems might get relegated toward the bottom of the list because of the time involved. It is more efficient to solve problems for a larger group of people than to fix this one user's problem immediately.

Some network administrators list all network service requests on a chalkboard or a whiteboard. They then prioritize them based on the previously discussed criteria. Some larger companies have written support-call tracking software whose only function is to track and prioritize all network and computer problems. Use whatever method makes you comfortable, but prioritize your calls.

Check the Software Configuration

Often, network problems can be traced to software configuration (as with our DNS configuration example earlier in this chapter). When you are checking for software problems, don't forget to check configuration, including the following:

- DNS configuration
- WINS configuration
- HOSTS file
- AUTOEXEC.BAT (DOS and Windows)
- CONFIG.SYS (DOS and Windows)
- STARTUP.NCF, AUTOEXEC.NCF, and server parameter settings (NetWare)
- The Registry (Windows 95/98 and NT)

Software configuration settings love to hide in places like these and can be notoriously hard to find (especially in the Registry).

Additionally, in text configuration files, look for lines that have been commented out (either intentionally or accidentally). A command such as REM or REMARK or the asterisk or semicolon characters indicate comment lines in a file.

Don't Overlook Physical Conditions

You want to make sure that from a network design standpoint, the physical environment for a server is optimized for placement, temperature, and humidity. When troubleshooting an obscure network problem, don't forget to check the physical conditions under which the network device is operating. Check for problems such as the following:

- Excessive heat
- Excessive humidity (condensation)
- Low humidity (leads to ESD problems)
- EMI/RFI problems
- ESD problems
- Power problems
- Unplugged cables

Don't Overlook Cable Problems

Cables, generally speaking, work fine once they are installed properly. Rarely is the cabling system the problem, unless someone has made some change to it. If you suspect that the cabling system is the problem, try replacing the patch cables at the workstation and hub first. These are easiest to

get to (and replace). If that solves the problem, you know the problem was related to the patch cable. It was either faulty or the wrong type.

If the patch cable isn't the problem, use a cable tester (not a tone generator and locator) to find the source of the problem. Wires that are moved can be prone to breaking or shorting. A short can happen when the wire conductor comes in contact with another conductive surface, changing the path of the electrical signal. The signal will go someplace else instead of to the intended recipient. You can use cable testers to test for many types of problems, including:

- Broken cables
- Incorrect connections
- Interference levels
- Total cable length (for length restrictions)
- Cable shorts
- Connector problems

Check for Viruses

Many troubleshooters overlook virus scanning because they assume that the network virus-checking software should have picked up any viruses. We're reminded by the network virus-scanning software of the bio-filters in the transporter on *Star Trek: The Next Generation*. They work great as long as the computer has the latest information on what the virus is and how to eliminate it. On many occasions, though, the ship's doctor or engineer had to reprogram the bio-filters to recognize some new virus that the crew of the *Enterprise* had come across.

The same thing happens with network virus-scanning software; to be effective, it must be kept up-to-date. Updates are made available almost daily.

If you are having strange, unusual, irreproducible problems with a workstation, try scanning it with an up-to-date virus scan utility. You'd be surprised how many times people have spent hours and hours troubleshooting a strange problem, only to run a virus scan utility, find and clean one or more viruses, and have the problem disappear.

Where can a network fail?

When you're trying to troubleshoot something as complex as a network, it's often best to take what you know about the problem and gradually narrow down the possible causes. With this in mind, we'll begin with simple problems and work toward more complicated ones. Basically, only four things can cause a network to fail:

- A problem with the server
- A problem with a hub, router, bridge, and so on
- A problem with the cabling
- A problem with a workstation
- A problem with NIC
- Protocol mismatches

Once you figure out which of these five items is causing your problem, you can begin narrowing your focus toward subcomponents.

For example, suppose you've determined that you have a cable problem. You've eliminated three quarters of your network components from the list of possible failures. Next, you'll want to investigate if you have a break in the cable, if an end is loose, if you have a missing terminator, or if you have some other problem. Let's examine the steps you'll follow in the troubleshooting process.

A starting point

The very first thing you need to find out is which PCs are affected. Are you having problems with one PC, several PCs, or every computer on the network?

Problems with a single PC

- ☞ If a problem is occurring only with a single PC, you've just narrowed the cause of your problem considerably. The next question you should ask is, "Has this PC ever worked before at this location?" If it has worked before and you know that nothing has changed in the configuration, then:
 - ✍ A component may be unplugged,
 - ✍ A cable may be broken, or
 - ✍ A network card may have gone out.
- ☞ On the other hand, if the PC is brand-new, chances are good that it's experiencing a configuration error. In a situation like this, your next step should be to plug a known good laptop computer into the network connection currently used by the failing PC.
 - ✍ If the laptop connects to the network successfully, you know there are no cable breaks or problems with the hubs or routers the problem is with the PC itself.
 - ✍ If the laptop also fails to connect, you've obviously got a cable problem or a problem with a switching device (such as a hub or router).

Problems with several PCs

If you're having problems with several PCs, you need to consider which PCs are malfunctioning.

- ☞ If they're all new, you've probably got a configuration problem. On the other hand, if all the PCs were working previously, you most likely have a problem with a switching device or a cable segment.
- ☞ At this point, you should determine what the failing PCs have in common with one another but not with the rest of the network. Many times, when multiple PCs fail simultaneously, a hub has gone bad or a cable connecting one hub to another hub has gone bad. You should check whether all the PCs share a common hub or are attached to the same cable segment.

Problems with the entire network

If no PCs on your network can log on, you've probably got problems with the server functioning as your primary domain controller. You should go to this server to begin the troubleshooting process.

Protocol mismatches

As you may know, in order for PCs on a network to communicate, they must share a common protocol. A protocol is a language that the computers on a network use to speak to one another. If you notice that a workstation can see only some of the other PCs on your network, there's a chance you may have a protocol mismatch.

To solve this problem, begin by double-clicking the Network icon in Control Panel. When you see the Network Properties dialog box, select the Protocols tab. Then, compare the protocols listed for this PC with the protocols installed on a PC it can't communicate with. If the protocols match, the PCs should be able to communicate with each other.

The NetBEUI protocol is an exception to this rule. NetBEUI isn't a routable protocol. If a router is located between two PCs using NetBEUI and you haven't made special provisions, the PCs won't be able to communicate. The easiest solution to this problem is to install NWLink IPX/SPX or TCP/IP.

A word about TCP/IP

With the increasing popularity of the Internet, TCP/IP is being used more often. Because of this growing usage, we'll show you how to troubleshoot some common TCP/IP problems.

If you suspect that you have a problem with TCP/IP, the first thing you should do is PING your own IP address. If the PING returns, then TCP/IP is functional. However, this means only that the protocol is functionality doesn't necessarily mean that TCP/IP can communicate with the rest of the world.

If the malfunctioning PC can successfully PING itself, you should PING the address of another PC on the same network segment as the malfunctioning PC. Doing so will test your network card. If the PING fails, the network card is malfunctioning.

If the PING is successful, PING the same PC using the PC's computer name rather than its IP address. If this PING fails, you probably have a problem with your WINS or DNS configuration, or possibly with your LMHosts file.

If you can successfully PING a PC by name, it's time to PING a PC that's either on a different network segment or in the outside world. For example, you might PING <http://www.xpressions.com>. If this PING fails but previous PINGs have been successful, you probably have either an incorrectly configured default gateway or a problem with your router.

However, if other PCs on the segment are functioning correctly, the problem is probably the default gateway rather than the router.

Connectivity testing with Ping, Telnet, Tracert and PathPing:

All of the following command line tools are accessed from the command prompt. You can open a command prompt window by selecting Start | All Programs | Accessories | Command Prompt.

You can also open the command prompt window by selecting Start | Run - and then entering CMD.EXE into the dialog box and pressing the Enter key or the OK button.

Each tool in this KB is given only a very basic overview and usage description. We would suggest that you research each of these in more detail to learn about advanced usage.

1. PING:

The ping command is a very simple connectivity testing tool. Ping verifies connectivity by sending Internet Control Message Protocol (ICMP) echo packets to a host and listening for an echo reply.

The ping command waits for each packet sent and prints the number of packets transmitted and received. Each received packet is validated against the sent packet. The default setting will send four echo packets containing 64 bytes of data. You can use the ping utility to test both the host name and IP address of the host for DNS resolution. A successful IP ping and failed host name ping could indicate name resolution issues.

Usage:

In a command prompt window, enter Ping followed by the Fully Qualified Domain Name (FQDN) or IP address of the server you want to test. You may wish to use the -t command line switch to send continuous echo requests to a host.

Ping 123.123.123.123

Common usage examples might be to test for a server to be restarted and start responding again. You may wish to use the `-t` command line switch to send continuous echo requests:

Ping 123.123.123.123 `-t`

Another example may be to test what IP address is returned by a specific record or service lookup:

Ping mail.domain_name.com

Ping www.domain_name.com

2. TELNET:

Telnet comes from the combination of the words telephone and network. It was originally designed to allow for command line remote management over slower connection types. RFC 854 states: “The purpose of the TELNET Protocol is to provide a fairly general, bi-directional, eight-bit byte orientated communications facility. “

It is a TCP based protocol that can also be used to test a variety of services for connectivity. You can use it to test for SMTP, SQL or Remote Desktop connectivity. This is a good test to use for service or port blocks resulting from a firewall configuration.

Usage:

In a command prompt, enter TELNET, followed by the Fully Qualified Domain Name (FQDN) or IP address of the server you want to connect to - and then the port that the service uses.

TELNET 123.123.123.123 5678

The following is a list of common protocols and ports of interest:

FTP	21
SMTP	25
SQL	1433
RDP	3389

The response of a successful connection will be different for each service, but a failed connection will always respond with a variation of the following message: "Could not open connection to the host, on port n: Connect failed"

When testing your mail connection with Telnet, you will want to reference the mail record for the domain:

```
TELNET mail.yourdomain.com 25
```

3. TRACERT:

Tracert is the Windows implementation of the traceroute tool that originated on UNIX and Cisco systems. Tracert is a Windows command-line tool that displays the path a packet takes to reach a destination from the machine that it is executed on. It does this by sending Internet Control Message Protocol (ICMP) echo request messages to the destination. It does this by incrementally increasing the Time To Live (TTL) values to find the path taken to the destination address. The path is displayed as a list in the order of which it heard back from each node that it passed through on its way to the destination.

When you run tracert, the top line shows the destination of the trace. It also lets us know that it stops if it reaches a maximum of 30 hops. Next you will see each hop it takes to reach the destination. The number of hops will go in order numerically from 1 to 30 depending on the path to the destination. Following this tracert will normally include at least 4 pieces of information for each hop; the number of the hop, the Round Trip Time (RTT is displayed in milliseconds or ms) it takes to get from the interface of the current hop and then back again to your machine, the IP

address of the interface for that hop and the hostname corresponding to the IP address of the hop. The default is to send out 3 packets to each hop. This is done in case a packet is lost and allows you to get an idea of whether or not there is a variance in the time for a specific hop.

A high number on the first external hop from your machine is a good indication of possible Local Area Network (LAN) issues.

An asterisk (*) indicates an echo request that was lost. These can be the result of security implementations of firewalls or Access Control List (ACL's). Additionally, routers may be configured not to respond to this type of traffic. You may see a row of three asterisks with no IP address or hostname. The trace may then continue responding normally again and display the destination results.

Usage:

In a command prompt window, enter TRACERT followed by the Fully Qualified Domain Name (FQDN) or IP address of the server you want to test. You may wish to use the `-d` command line switch to prevent Tracert from resolving the name of the nodes from the IP address in the trace route.

TRACERT 123.123.123.123

You can see the output results in the following example:

Tracing route to 123.123.123.123 over a maximum of 30 hops:

```

1  <1 ms  <1 ms  <1 ms  111.111.111.111
2  <1 ms  <1 ms  <1 ms  222.222.222.222
3   1 ms   1 ms   1 ms  111.222.111.222
4   1 ms   1 ms  12 ms  222.111.222.111
5   *      *      *      Request timed out.
```

```
6  14 ms  13 ms  13 ms  123.123.123.123
```

Trace complete.

Additionally, you can use external tools such as www.traceroute.org or other ‘looking glass’ type sites to verify traces from different geographic locations throughout the world. You may want to select multiple sites to test connectivity to your server.

4. PATHPING:

PathPing is a utility that combines many of the features of Ping and Tracert into one tool. You can use it to verify connectivity to a host as well as see if you are taking an optimal path to a remote host or suffering from a bottleneck somewhere in the connection route. The final output provides statistics on the latency (packet loss) by sending multiple echo requests over a period of time to each node between the local and remote host.

Initially, PathPing will produce results are similar to Tracert; you will see the hop number followed by the IP address or node name. PathPing will then compute the statistics (the time this takes depends on the number of hops) for each node in the connection route. After the computation is complete, the window will display the following information for each node: Hop number, Round Trip Time (RTT), percent of packets Lost and Sent for Source to Here, the Address of the node at that hop and the percent of packets Lost and Sent from This Node/Link to the next node. You can see the output results in the following example:

		Source to Here		This Node/Link	
Hop	RTT	Lost/Sent = Pct	Lost/Sent = Pct	Address (Node)	
0				111.111.111.111/	
			0/100 = 0%		
1	30ms	0/100 = 0%	0/100 = 0%	222.222.222.222/	
			0/100 = 0%		

2	30ms	0/100 = 0%	0/100 = 0%	111.222.111.222/
			33/100 = 33%	
3	30ms	0/100 = 0%	0/100 = 0%	222.111.222.111/
			0/100 = 0%	
0	30ms	0/100 = 0%	0/100 = 0%	123.123.123.123

Trace complete.

The “Source to Here” – is the first set of statistic after the hop number is equivalent to if you pinged the node directly.

The “This Node/Link” is the set of statistic before the pipe and is the column you want to pay the most attention to. This will show you the statistics for the links between the nodes.

In the above example, the link between 111.222.111.222 and 222.111.222.111 is dropping 33 percent of the packets. The router at hop 3 is dropping packets addressed to it, but this loss does not affect their ability to forward traffic.

A 0/100 = 0% means that out of 100 packets, none were lost. A low single digit loss 1% or 2% is common, but anything higher is an indication of

Usage:

In a command prompt window, enter PathPing followed by the Fully Qualified Domain Name (FQDN) or IP address of the server you want to test. You may wish to use the `-n` command line switch to prevent PathPing from resolving name from the IP address of the nodes in the connection route.

`PATHPING -n 123.123.123.123`

PathPing offers slightly more accurate output over Tracert because it provides averages based on multiple echo requests. One disadvantage to PathPing is that it can take longer to return results.

ipconfig

In computing, ipconfig (internet protocol configuration) in Microsoft Windows is a console application that displays all current TCP/IP network configuration values and can modify Dynamic Host Configuration Protocol DHCP and Domain Name System DNS settings.

In most cases, the ipconfig command is used with the command-line switch /all. This results in more detailed information than ipconfig alone.

Syntax

```
ipconfig [/all] [/renew [Adapter]] [/release [Adapter]] [/flushdns] [/displaydns] [/registerdns]
[/showclassid Adapter] [/setclassid Adapter [ClassID]]
```

Parameters

- **/all** : Displays the full TCP/IP configuration for all adapters. Without this parameter, ipconfig displays only the IP address, subnet mask, and default gateway values for each adapter. Adapters can represent physical interfaces, such as installed network adapters, or logical interfaces, such as dial-up connections.
- **/renew [Adapter]** : Renews DHCP configuration for all adapters (if an adapter is not specified) or for a specific adapter if the Adapter parameter is included. This parameter is available only on computers with adapters that are configured to obtain an IP address automatically. To specify an adapter name, type the adapter name that appears when you use ipconfig without parameters.
- **/release [Adapter]** : Sends a DHCPRELEASE message to the DHCP server to release the current DHCP configuration and discard the IP address configuration for either all adapters

(if an adapter is not specified) or for a specific adapter if the Adapter parameter is included. This parameter disables TCP/IP for adapters configured to obtain an IP address automatically. To specify an adapter name, type the adapter name that appears when you use ipconfig without parameters.

- **/flushdns** : Flushes and resets the contents of the DNS client resolver cache. During DNS troubleshooting, you can use this procedure to discard negative cache entries from the cache, as well as any other entries that have been added dynamically.
- **/displaydns** : Displays the contents of the DNS client resolver cache, which includes both entries preloaded from the local Hosts file and any recently obtained resource records for name queries resolved by the computer. The DNS Client service uses this information to resolve frequently queried names quickly, before querying its configured DNS servers.
- **/registerdns** : Initiates manual dynamic registration for the DNS names and IP addresses that are configured at a computer. You can use this parameter to troubleshoot a failed DNS name registration or resolve a dynamic update problem between a client and the DNS server without rebooting the client computer. The DNS settings in the advanced properties of the TCP/IP protocol determine which names are registered in DNS.
- **/showclassid Adapter**: Displays the DHCP class ID for a specified adapter. To see the DHCP class ID for all adapters, use the asterisk (*) wildcard character in place of Adapter. This parameter is available only on computers with adapters that are configured to obtain an IP address automatically.
- **/setclassid Adapter [ClassID]**: Configures the DHCP class ID for a specified adapter. To set the DHCP class ID for all adapters, use the asterisk (*) wildcard character in place of Adapter. This parameter is available only on computers with adapters that are configured to obtain an IP address automatically. If a DHCP class ID is not specified, the current class ID is removed.

- `/?`: Displays help at the command prompt.

Final note:

All of these Windows utilities are based on ICMP echo request over TCP/IP - otherwise known as ping packets. Many firewalls block ICMP traffic - so you may not get the response although the site is up and responsive. Access rules can cause false negatives with the reporting of from these network tools.